



北京大学
PEKING UNIVERSITY

第六章 固体理论

1. 晶体结构

物质的四种基本形态：气、液、固、等离子体

	气体	液体	固体
粒子间距离	可任意大	基本是常数	较小，~键长
密度	可任意小	基本是常数	较大
是否能被容器塑形	是	是	否
是否一定充满容器	是	否	否
是否容易压缩	容易	难	极难，高压可以

兵无常势，水无常形

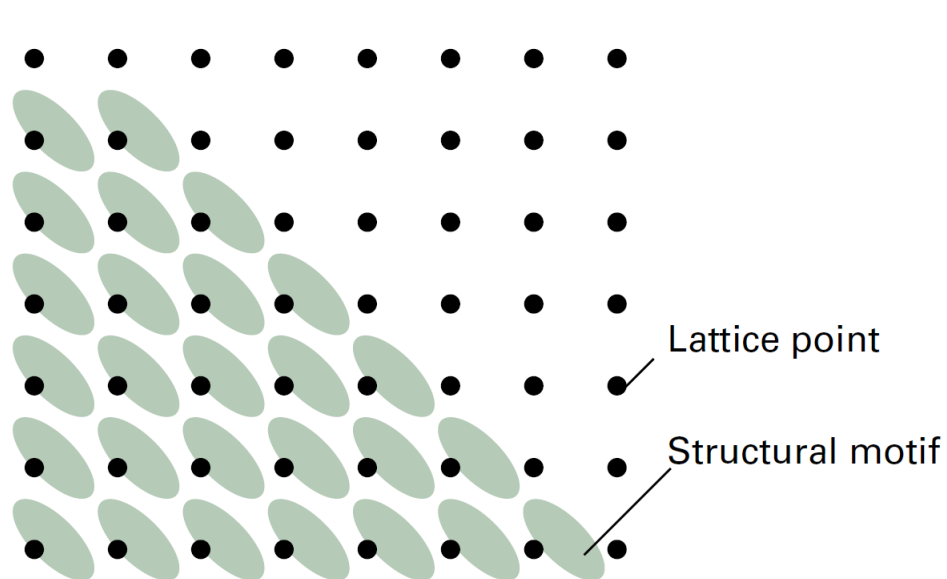
无孔不入 水泄不通

固体：晶体、无定形固体

晶体：周期性排列

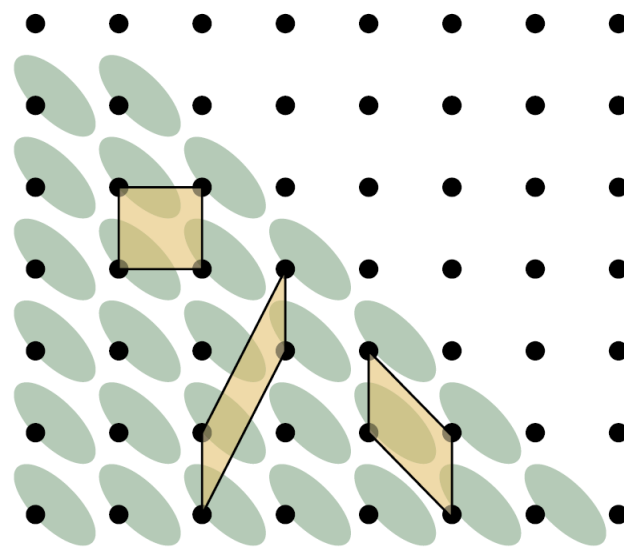
无定形固体：非周期性排列

晶体结构 = 点阵 + 结构基元

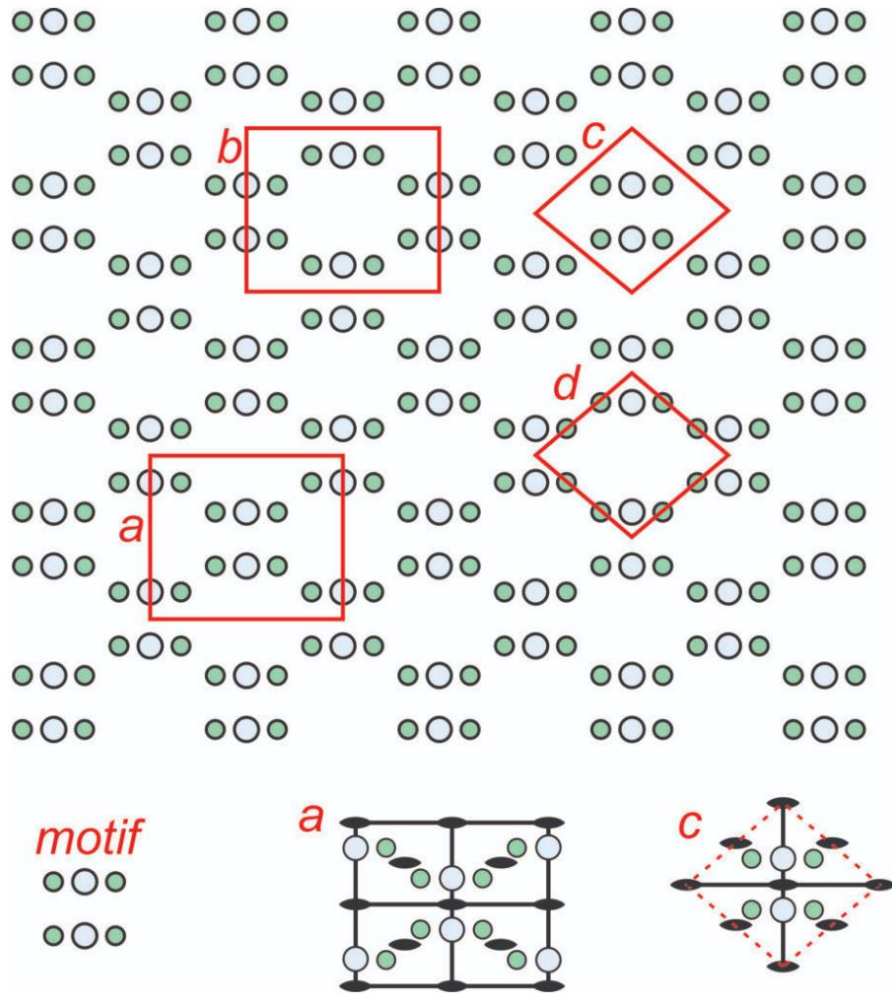


晶胞 (unit cell)：特定的平行多面体，通过平移能覆盖整个点阵

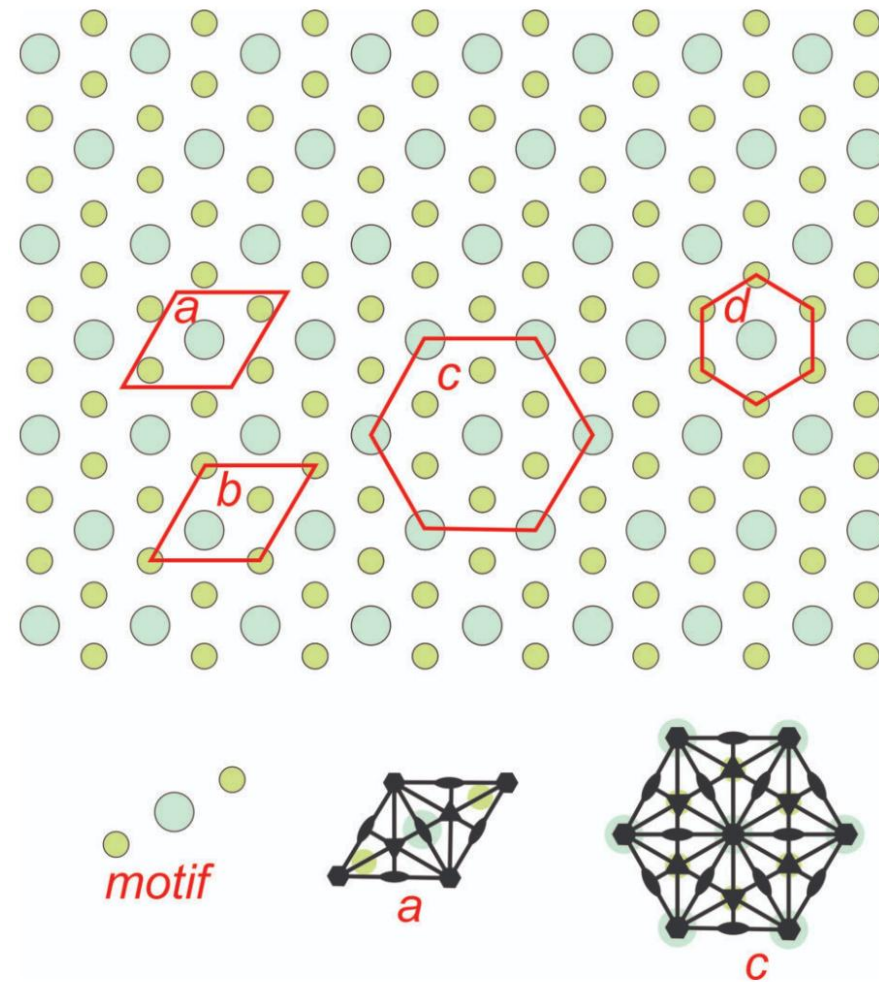
二维：平行四边形 } 一般是这样
三维：平行六面体 }



注意区分结构基元与晶胞



11.21 Motifs and unit cells with orthorhombic symmetry



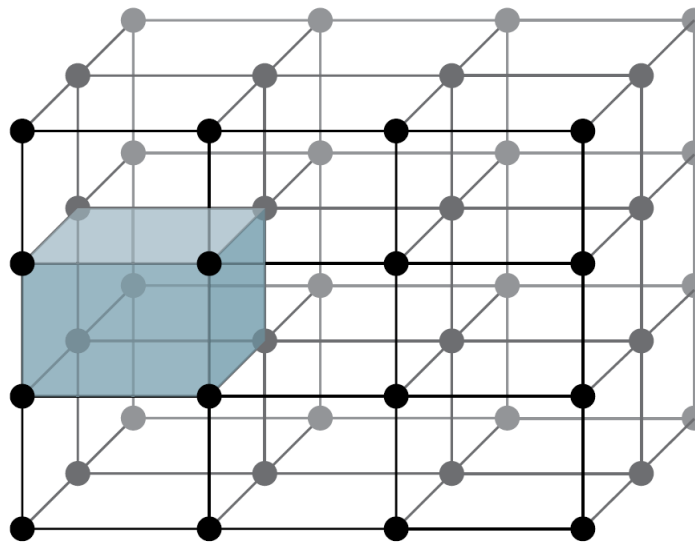
11.22 Motif's and unit cells with rhombohedral or hexagonal symmetry

素晶胞 (primitive cell) :

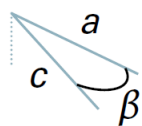
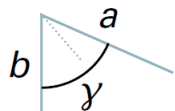
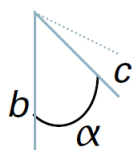
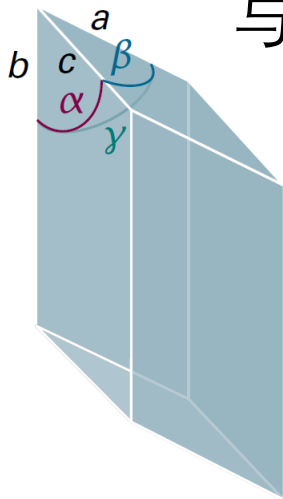
只包含一个点阵点

复晶胞 (non-primitive) :

包含多个点阵点



平行六面体的三个边长
与三个角度



选取晶胞的方式有无数种，但一般选取边长小的，角度接近直角的

问题一 晶体形貌与晶系

晶体的对称性（对称元素）：

- 镜面
- 反演中心
- 旋转轴

{ 普通旋转轴 C_n
 rotation axis
 旋转反演轴（反轴） I_n
 rotoinversion axis

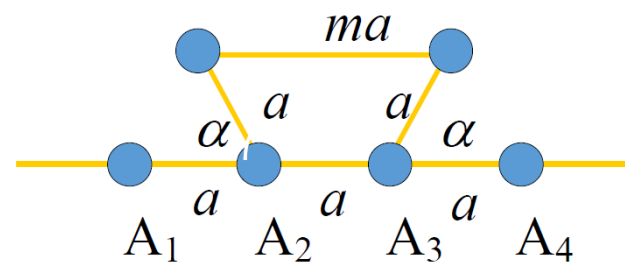
$$I_n^1 = iC_n^1$$

复习：映轴 S_n

$$S_n^1 = \sigma C_n^1$$

普通旋转轴只有2, 3, 4, 6重轴

受限于平移对称性



$$a + 2a\cos\alpha = ma$$

$$\cos\alpha = (m - 1) / 2$$

$$| (m - 1) / 2 | \leq 1$$

m	-1	0	1	2	3
cos α	-1	-0.5	0	0.5	1
α	180°	120°	90°	60°	0°
n	2	3	4	6	1

晶体学 (crystallography) 对称元素记号

- 镜面 m
- 反演中心 i
- 旋转轴: 1, 2, 3, 4, 6
- 反轴: $\bar{1}$, $\bar{2}$, $\bar{3}$, $\bar{4}$, $\bar{6}$
- 旋转轴+镜面: $1/m$, $2/m$, $3/m$, $4/m$, $6/m$

共13种不同对称元素

i
 $m = \bar{2} = 1/m$
1
2
3
4
6
 $\bar{3}$
 $\bar{4}$
 $\bar{6} = 3/m$
 $2/m$
 $4/m$
 $6/m$

Hermann-Mauguin
(国际) 符号 (自学)

如622, $4/m\bar{3}2/m$

三个轴的方向有规律

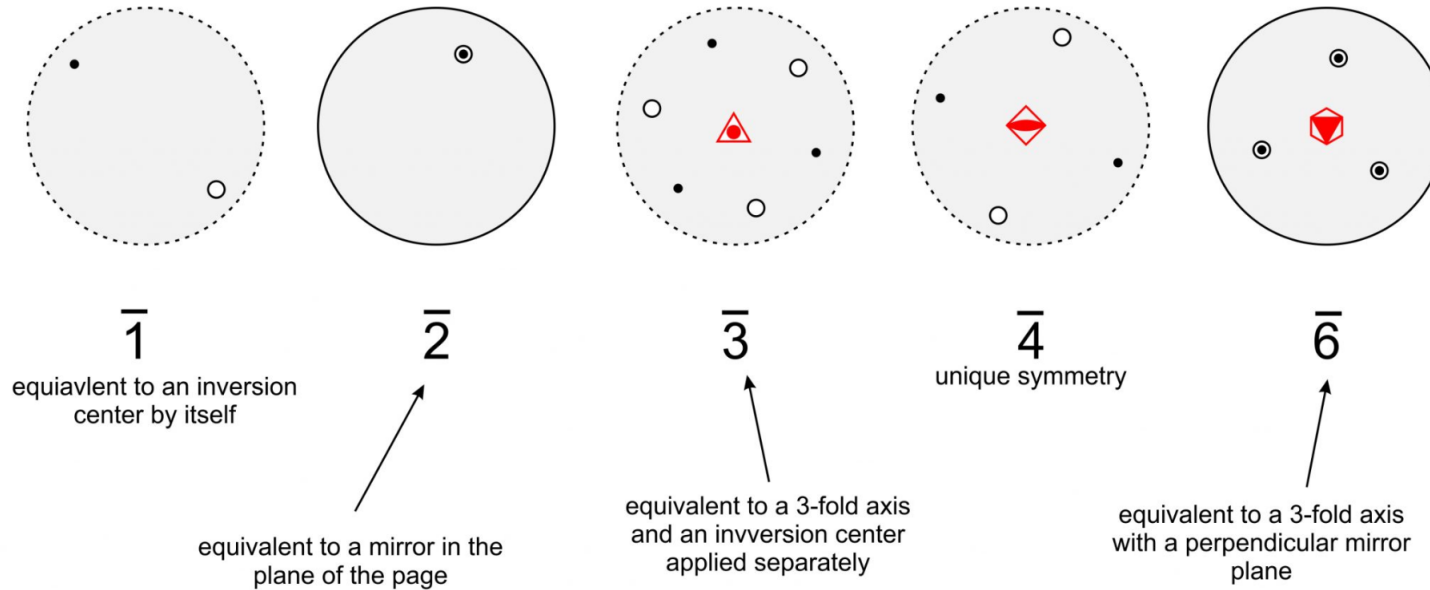
crystal system	possible first symbol	possible second symbol	possible third symbol	example of a point group
cubic	$4, 4/m, \bar{4}, 2, 2/m$	$3, \bar{3}$	$2, 2/m, m$	$\bar{4}3m$
hexagonal	$6, 6/m, \bar{6}$	$2, 2/m, m$	$2, 2/m, m$	$6/m^2/m^2/m$
trigonal	$3, \bar{3}$	$2, 2/m, m$	$2, 2/m, m$	32
tetragonal	$4, 4/m, \bar{4}$	$2, 2/m, m$	$2, 2/m, m$	422
orthorhombic	$2, m, 2/m$	$2, 2/m, m$	$2, 2/m$	$mm2$
monoclinic	$2, 2/m, m$			$2/m$
triclinic	$1, \bar{1}$			$\bar{1}$

表 7.4.1 国际记号中 3 个位置代表的方向

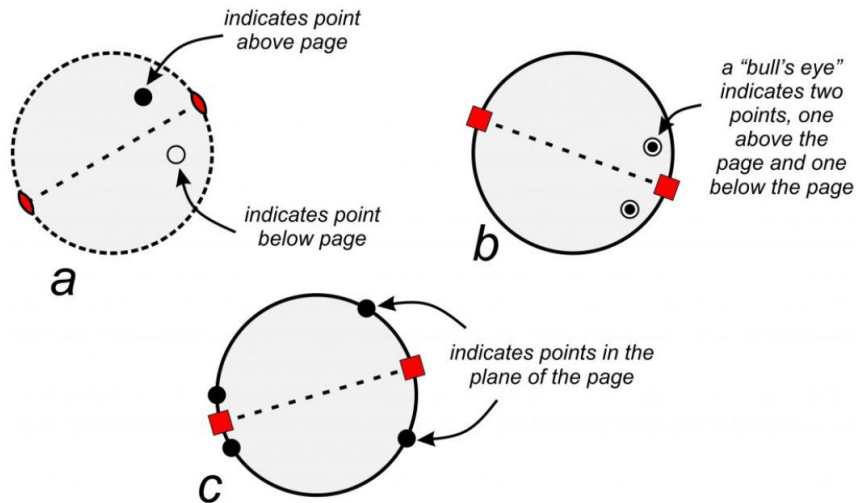
晶 系	3 个位置所代表的方向		
	1	2	3
立方晶系	a	$a+b+c$	$a+b$
六方晶系	c	a	$2a+b$
四方晶系	c	a	$a+b$
三方晶系(取菱面体晶胞)	$a+b+c$	$a-b$	—
三方晶系(取六方晶胞)	c	a	—
正交晶系	a	b	c
单斜晶系	b	—	—
三斜晶系	—	—	—

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Stereo diagrams



$$\bar{6} = 3/m$$



10.20 Rotation axes in the plane of the page

<https://opengeology.org/Mineralogy/11-crystallography/>
 Dexter Perkins, Mineralogy, 3rd edition, Pearson, 2011.

证明: $\bar{6} = 3/m$

等价于证明: $\bar{6} = C_{3h}$

C_{3h} 点群元素: $E, C_3^1, C_3^2, \sigma_h, S_3^1, S_3^5$

$$C_{3h} = C_3 \times C_s = \{E, C_3^1, C_3^2\} \times \{E, \sigma_h\}$$

$\bar{6}$ 点群元素: $E, \bar{6}, \bar{6}^2, \bar{6}^3, \bar{6}^4, \bar{6}^5$

$$\bar{6} = C_6^1 i = C_6^1 C_2^1 \sigma_h = C_3^2 \sigma_h = (C_3^1)^5 (\sigma_h)^5 = (C_3^1 \sigma_h)^5 = S_3^5$$

$$\bar{6}^2 = (C_6^1 i)^2 = (C_6^1)^2 i^2 = C_3^1$$

$$\bar{6}^3 = (C_6^1 i)^3 = (C_6^1)^3 i^3 = C_2^1 i = \sigma_h$$

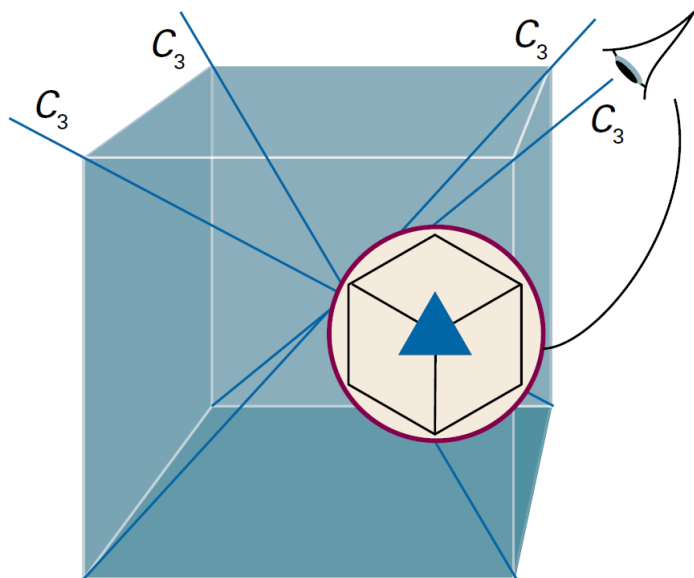
$$\bar{6}^4 = (\bar{6}^2)^{-1} = (C_3^1)^{-1} = C_3^2 \quad \bar{6}^5 = (\bar{6})^{-1} = (S_3^5)^{-1} = S_3^1$$

利用如下性质:

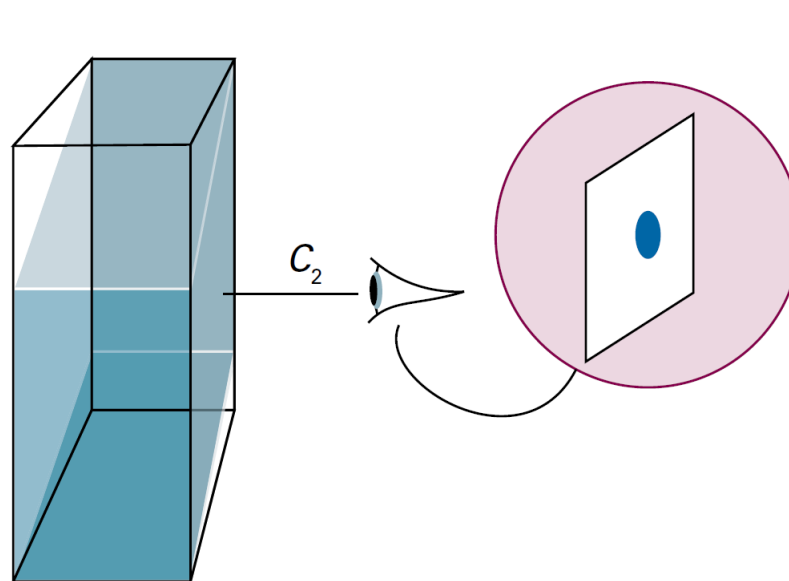
- 旋转与 i, σ_h 对易
- $i = C_2^1 \sigma_h$
- $\sigma_h = C_2^1 i$

晶体的分类

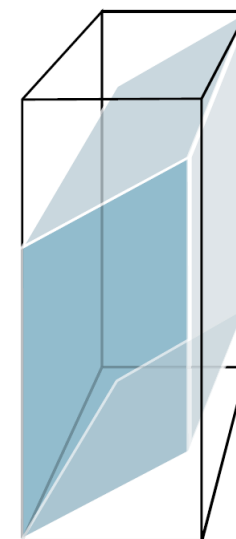
七种晶系（按照晶胞的旋转对称性划分）



立方晶系（4个 C_3 轴）



单斜晶系（1个 C_2 轴）



三斜晶系（无对称轴）

七种晶系及主要对称性

System		Essential symmetries
Triclinic	三斜	None
Monoclinic	单斜	One C_2 axis
Orthorhombic	正交	Three perpendicular C_2 axes
Rhombohedral	菱方	One C_3 axis 有时又叫trigonal 三方
Tetragonal	四方	One C_4 axis
Hexagonal	六方	One C_6 axis
Cubic	立方	Four C_3 axes in a tetrahedral arrangement

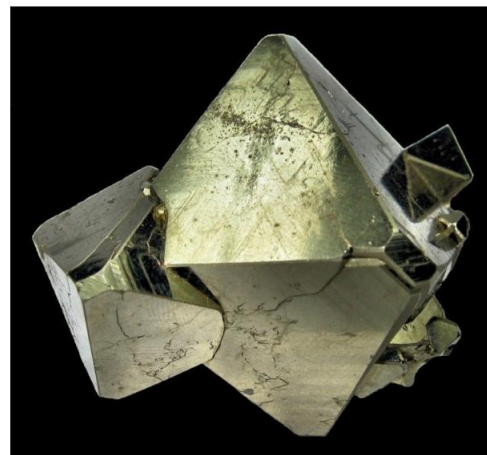
注：此表中的轴可以是旋转轴或反轴



10.48 Pyrope (garnet) ($\text{Mg}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) crystal with dodecahedral faces



10.49 Pyrite (FeS_2) pyritohedron. 3.2 cm across.

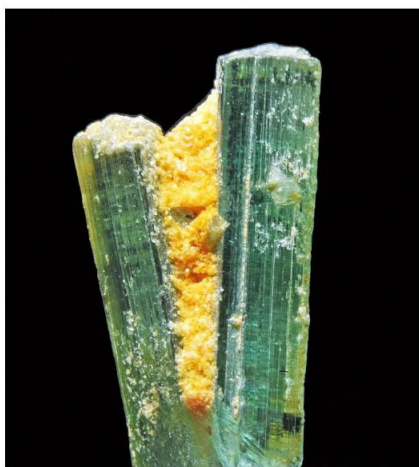


10.50 Pyrite (FeS_2) octahedra with small cubic faces

立方晶系



10.51 Hanksite (Na-Ca sulfate carbonate) from Searles Lake, California



10.52 Greenish tourmaline crystals from Paraiba, Brazil



10.53 Gray calcite (CaCO_3) with chalcopryite (CuFeS_2). The specimen is 12.5 cm wide.

六方、三方晶系



10.54 Green apophyllite on top of stilbite (Na-Ca zeolite). The specimen is about 16 cm tall.

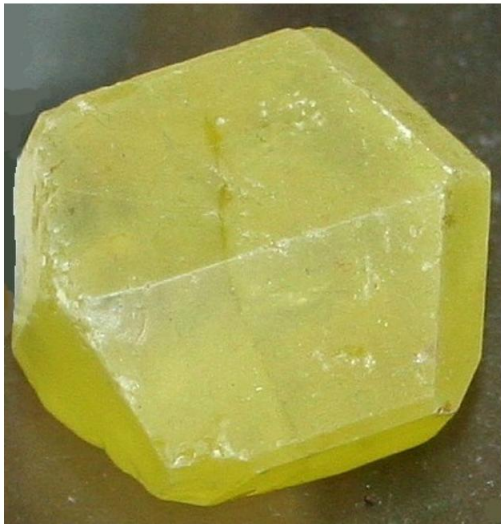


10.55 Vesuvianite (complex Ca-Mg-Fe silicate). The specimen is 4.4 cm wide.



10.56 Orange wulfenite (PbMoO_4) on top of calcite. Los Lamentos Mine, Mexico.

四方晶系



10.57 Sulfur (S) crystal



10.58 Topaz $\text{Al}_2\text{SiO}_4(\text{F},\text{OH})_2$. The crystal is 9.1 cm tall.



10.59 Orthorhombic prisms of barite (BaSO_4) terminated by pyramids.

正交晶系

单斜晶系



10.60 Clear crystal of gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)



10.61 A crystal of orthoclase KAlSi_3O_8 .
The crystal is 3 cm long.



10.62 Heulandite The specimen is 7.5 cm tall.

三斜晶系



10.63 Albite ($\text{NaAlSi}_3\text{O}_8$) crystal with minor tourmaline and quartz



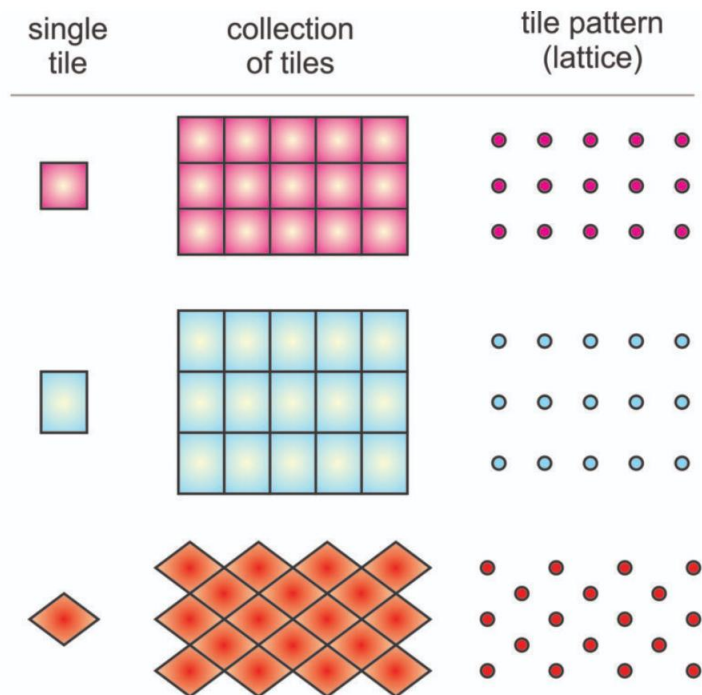
10.64 Amblygonite (an Li-Al phosphate) from Minas Gerais, Brazil



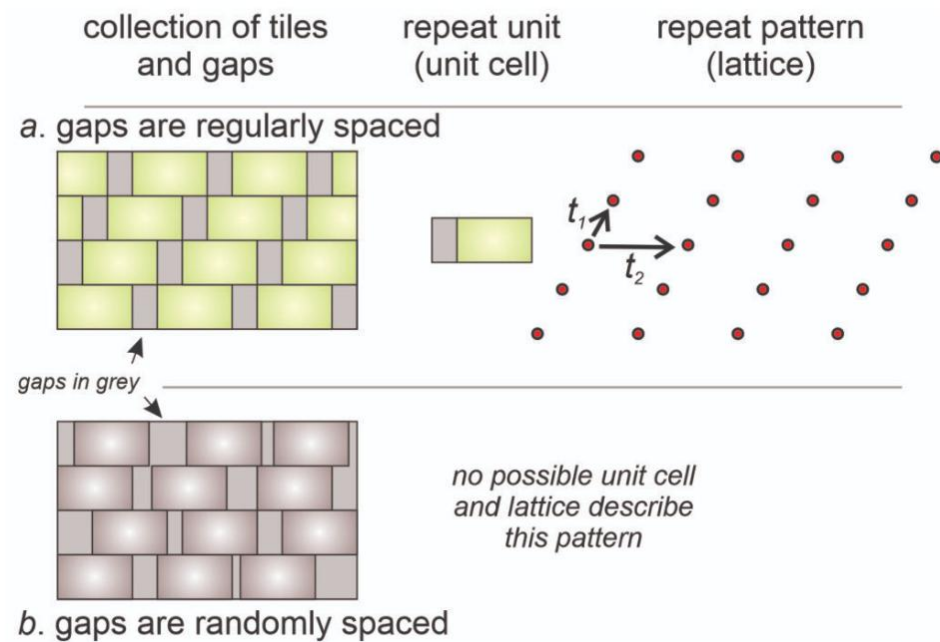
10.65 Rhodonite (MnSiO_3). 5.7 cm tall

问题二 点阵、点群、 空间群

二维点阵（铺地板）



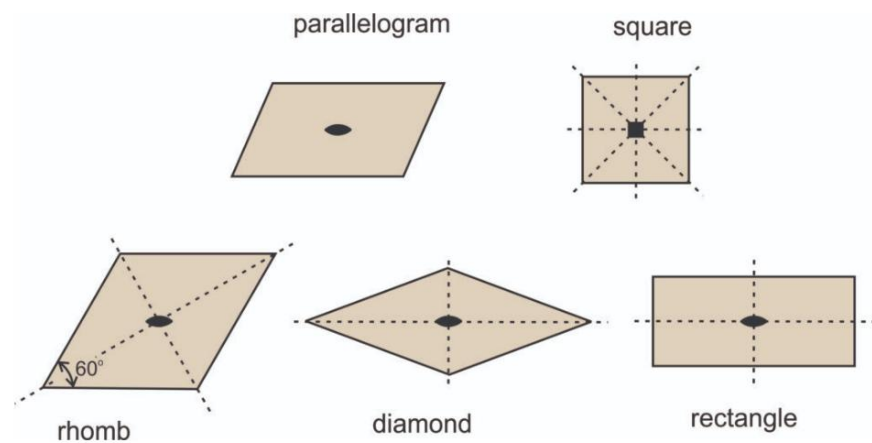
铺地砖的几种方式



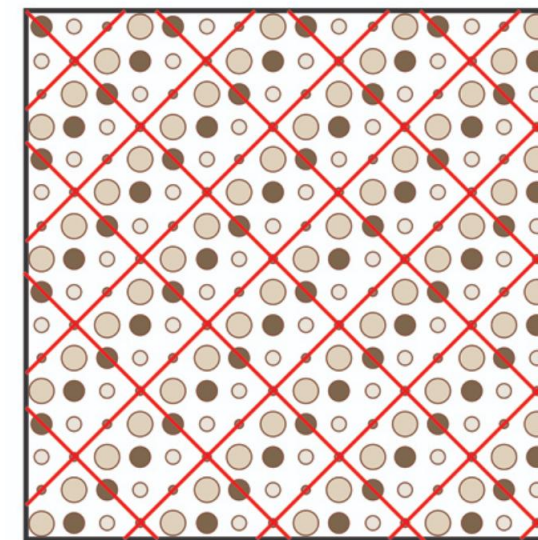
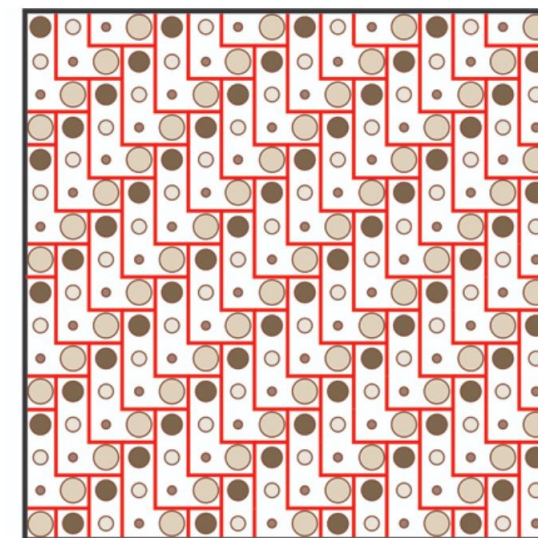
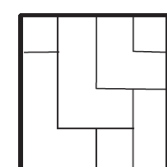
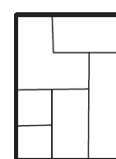
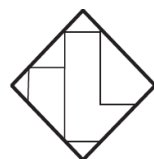
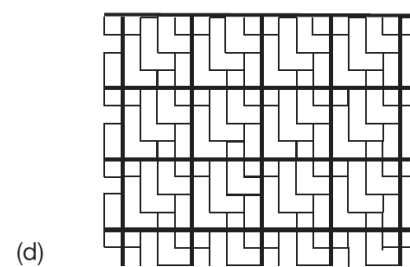
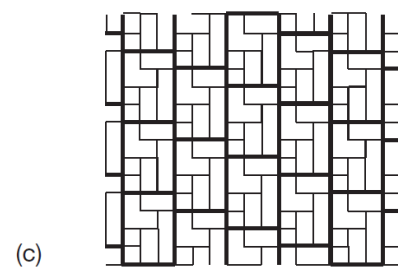
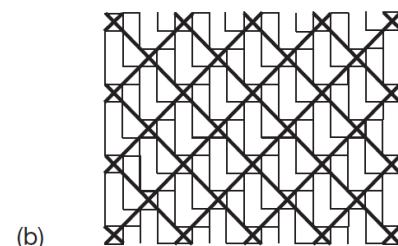
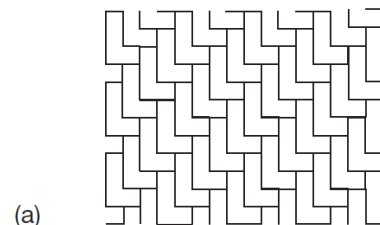
通过平移须铺满平面

水泥缝隙也要有平移对称性

无论图案多复杂，都可以
抽出**五种基本形状（晶胞）**

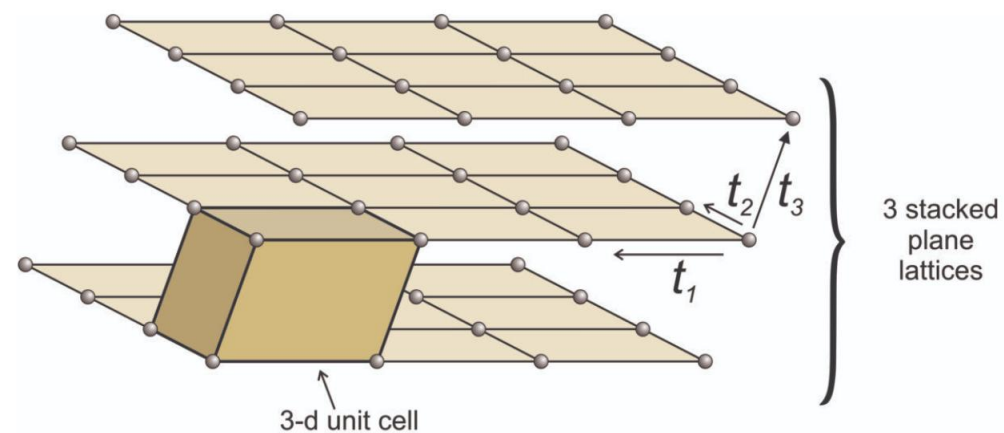


五种平面点阵
(plane lattice)

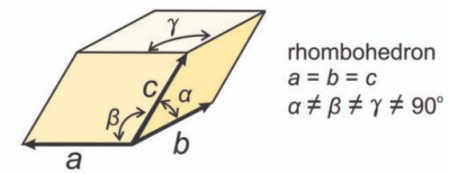
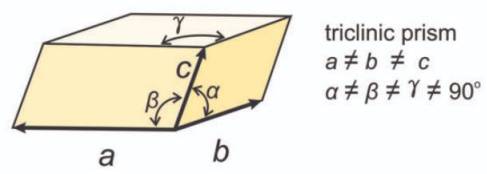
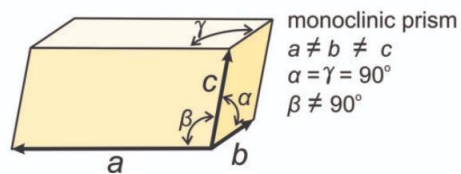
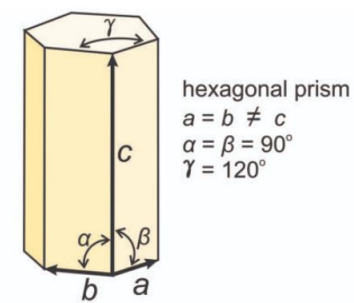
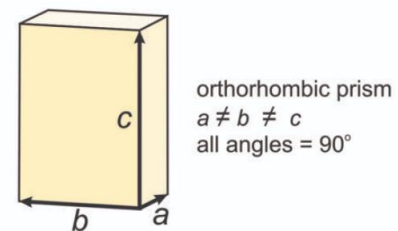
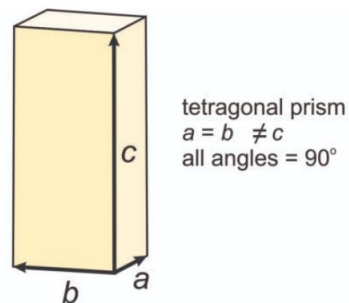
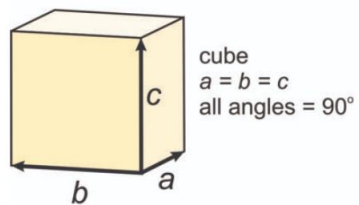


三维点阵的单胞

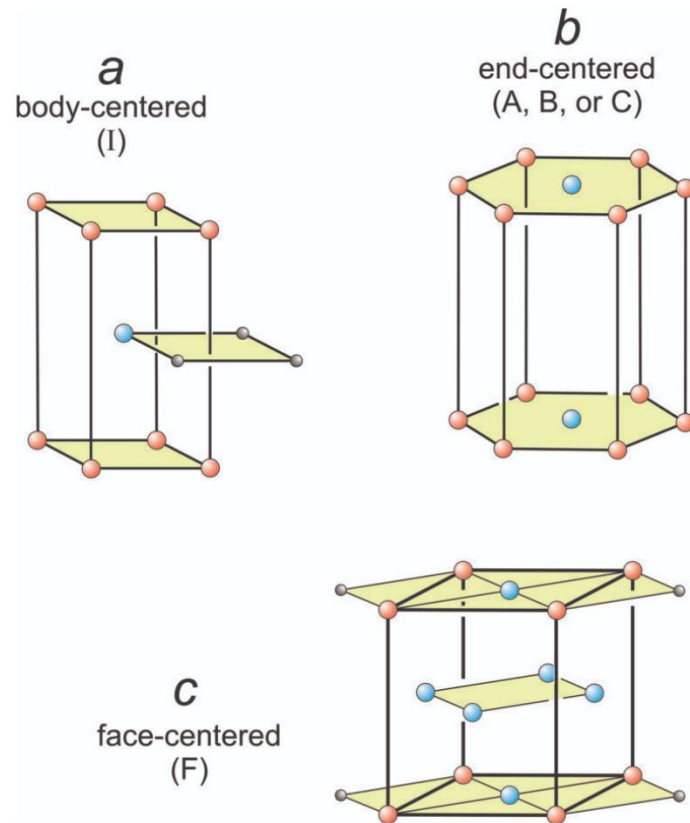
平面点阵 + 第三维 = 平面点阵空间堆叠



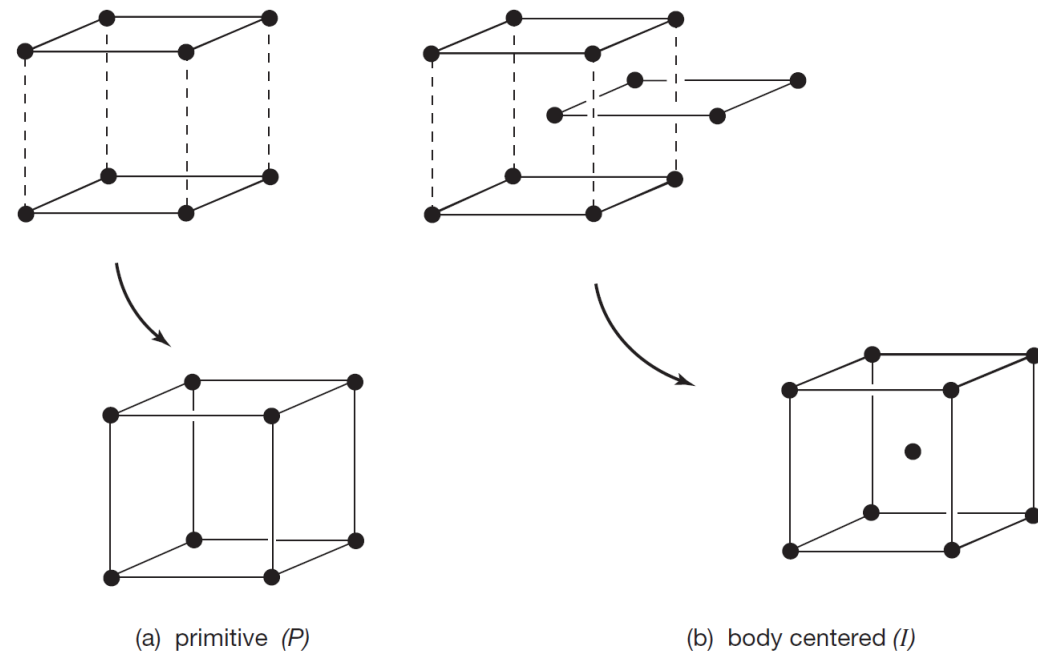
七种素晶胞



平面点阵空间堆叠还可得到复晶胞



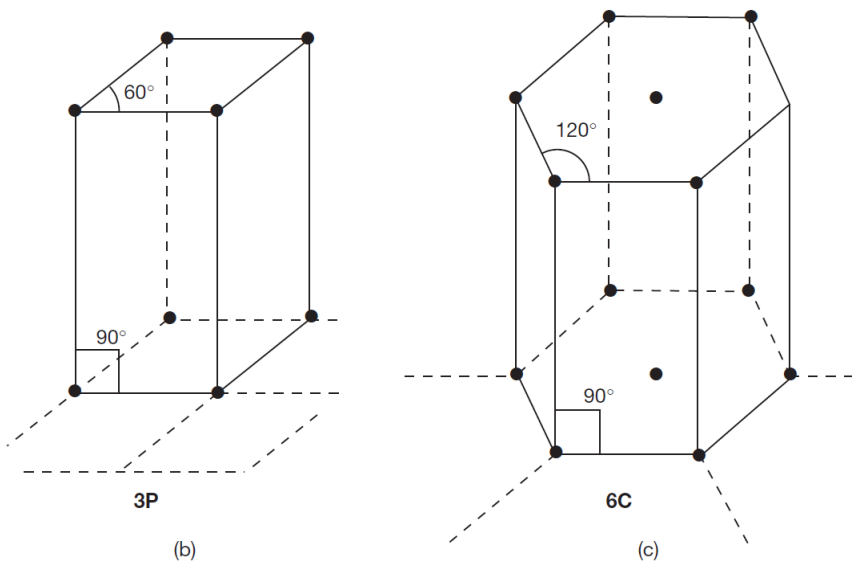
错位堆叠



六方平面点阵的两种堆叠方式

1. 保留六重轴（两层重叠）

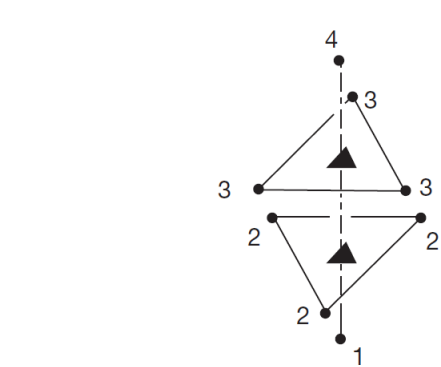
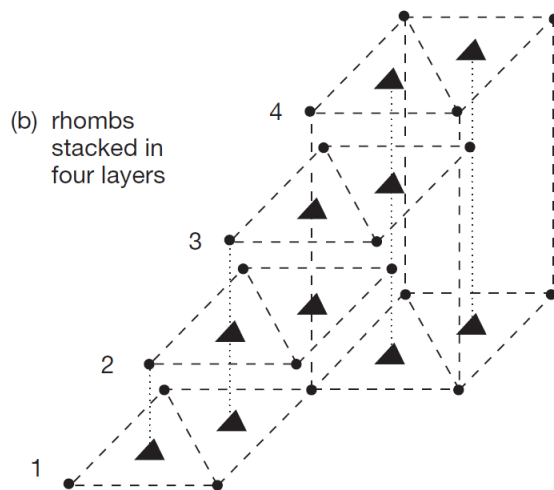
2. 降成三重轴（三层交错）



六方晶系

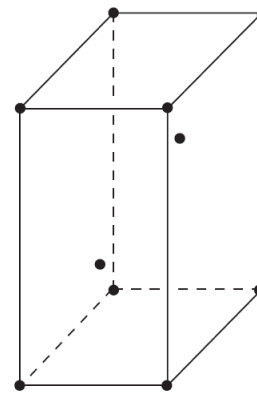


(a) rhomb and rotation axes



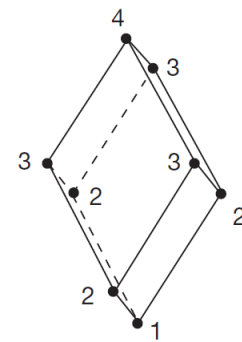
(d) lattice points needed to make a rhombohedron

注意素晶胞与复晶胞



(c) nonprimitive rhombohedral unit cell (R)

三方晶系



(e) primitive rhombohedral unit cell (R)

考虑三维全部的堆叠方式

14种布拉维点阵 (Bravais lattices)

记号含义:

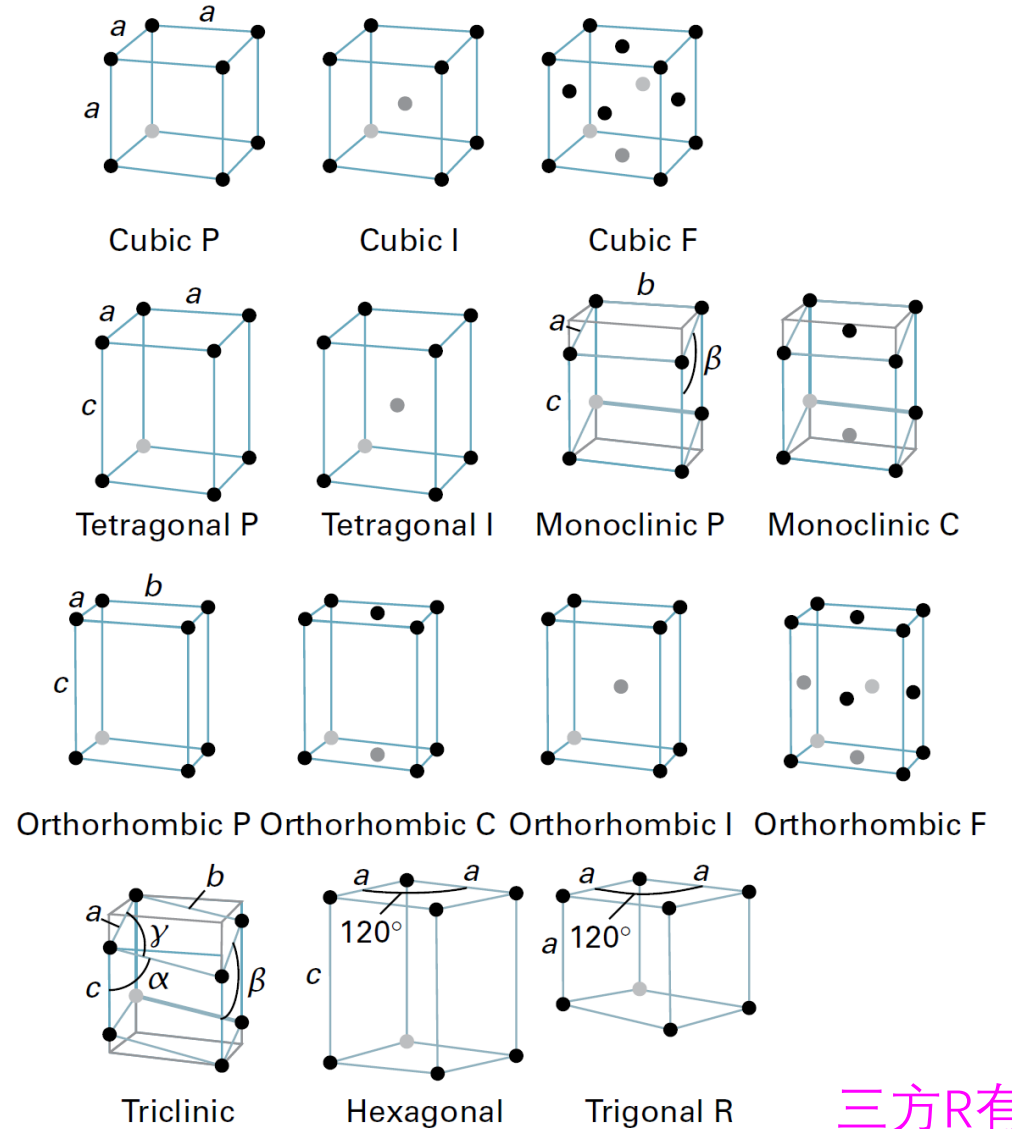
P: primitive素晶胞

I: innenzentrierte体心占据

F: face-centered面心占据

C: 边心 (底心) 占据

R: rhombohedral



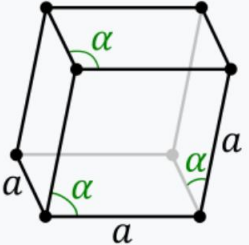
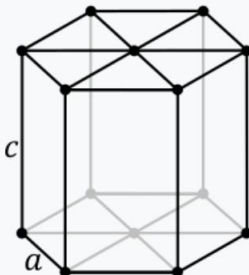
三方R有误

一些过时概念：

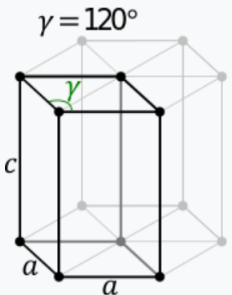
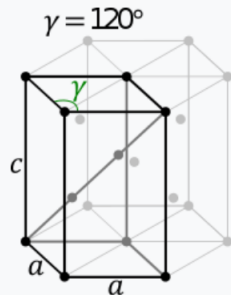
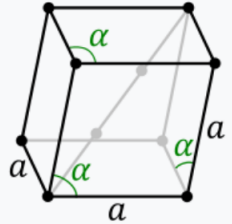
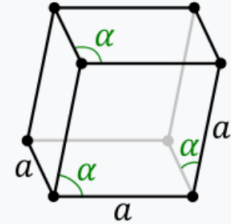
点阵系统、晶族

注意区分概念：晶系、点阵、

晶胞、素（复）晶胞

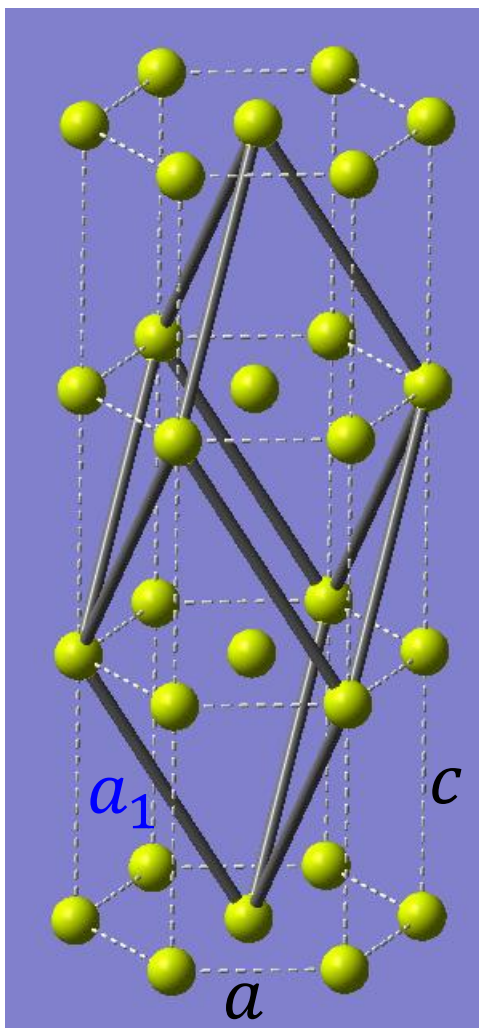
Crystal system	Trigonal		Hexagonal
Lattice system	 Rhombohedral	 Hexagonal	
Example	 Dolomite	 α-Quartz	 Beryl

Hexagonal crystal family

Bravais lattice	Hexagonal	Rhombohedral
Pearson symbol	hP	hR
Hexagonal unit cell		
Rhombohedral unit cell		

简单六方(hP)与菱方(hR)点阵分别都有两种晶胞取法

https://en.wikipedia.org/wiki/Hexagonal_crystal_family

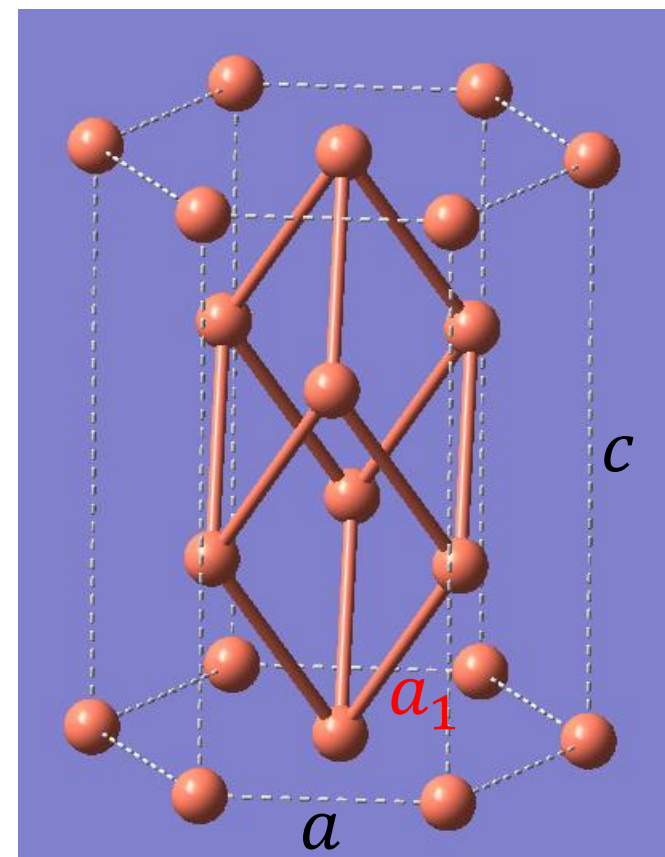


hP点阵中取菱方晶胞

Hexagonal crystal family		
Bravais lattice	Hexagonal	Rhombohedral
Pearson symbol	hP	hR
Hexagonal unit cell	$\gamma = 120^\circ$ 	$\gamma = 120^\circ$
Rhombohedral unit cell		

$$a_1 = \sqrt{a^2 + c^2}$$

$$\cos \alpha = 1 - \frac{3}{2(1 + \frac{c^2}{a^2})}$$



hR点阵中取菱方晶胞

$$a_1 = \frac{\sqrt{3a^2 + c^2}}{3}$$

$$\cos \alpha = 1 - \frac{9}{2(3 + \frac{c^2}{a^2})}$$

晶胞对称性与晶体对称性

很多晶胞堆在一起构成晶体

晶体对称性不可能高于晶胞

研究一个晶胞的对称性

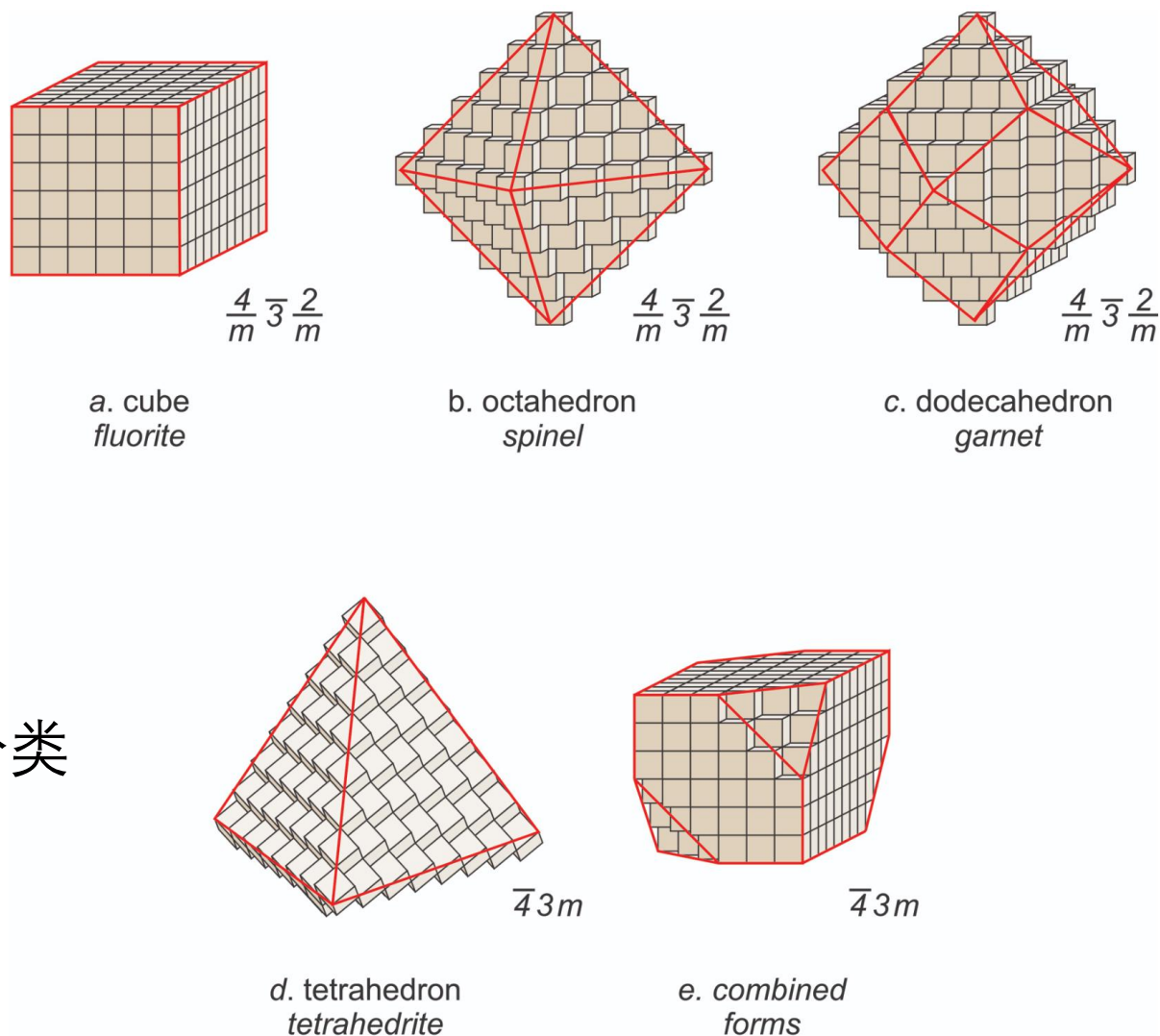
= 研究结构基元的对称性

只考虑点操作对结构基元进行分类

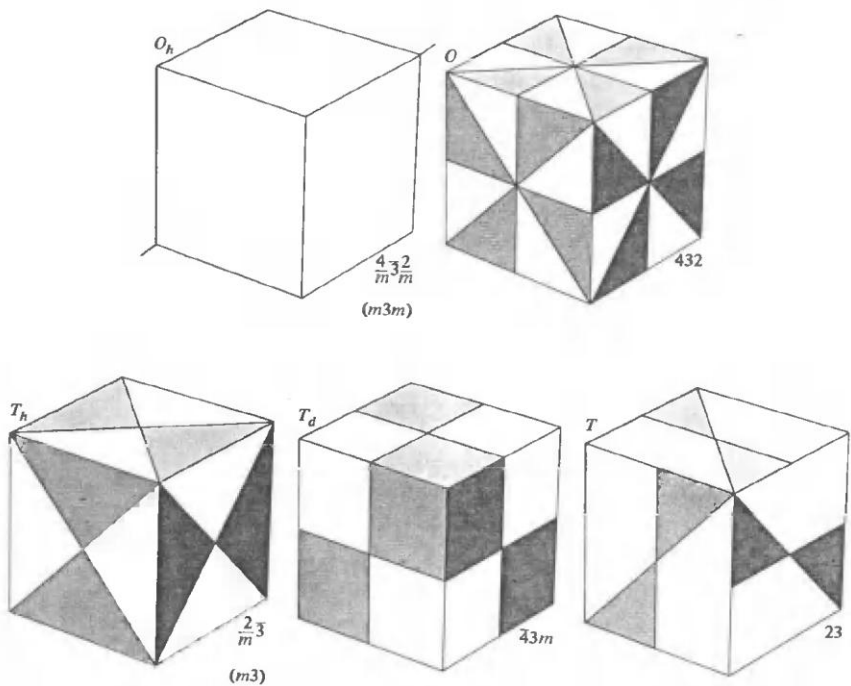
不再抽象成点阵点



晶体学点群



32种晶体学点群



立方晶分成五类

比较两种符号（Schönflies vs Hermann Mauguin）

Crystal Systems and Point Groups		
crystal system	unit cell shape	point groups
triclinic	triclinic prism	$1, \bar{1}$
monoclinic	monoclinic prims	$2, m, ^2/m$
orthorhombic	orthorhombic prism	$222, mm2, mmm$
tetragonal	tetragonal prism	$4, \bar{4}, ^4/m, 422, 4mm, \bar{4}2m, ^4/m^2/m^2/m$
trigonal	rhombohedron or hexagonal prism	$3, \bar{3}, 32, 3m, \bar{3}^2/m$
hexagonal	hexagonal prism	$6, \bar{6}, ^6/m, 622, 6mm, \bar{6}2m, ^6/m^2/m^2/m$
cubic	cube	$23, ^2/m\bar{3}, 432, \bar{4}3m, ^4/m\bar{3}^2/m$



7大晶系、14个点阵、32个点群、230个空间群的联系

	BRAVAIS LATTICE (BASIS OF SPHERICAL SYMMETRY)	CRYSTAL STRUCTURE (BASIS OF ARBITRARY SYMMETRY)
Number of point groups:	7 ("the 7 crystal systems")	32 ("the 32 crystallographic point groups")
Number of space groups:	14 ("the 14 Bravais lattices")	230 ("the 230 space groups")

晶体可以平移+旋转



230个空间群

思考：为什么不是
 $14 \times 32 = 448$ 种？

不考虑平移，让至
少一个点固定

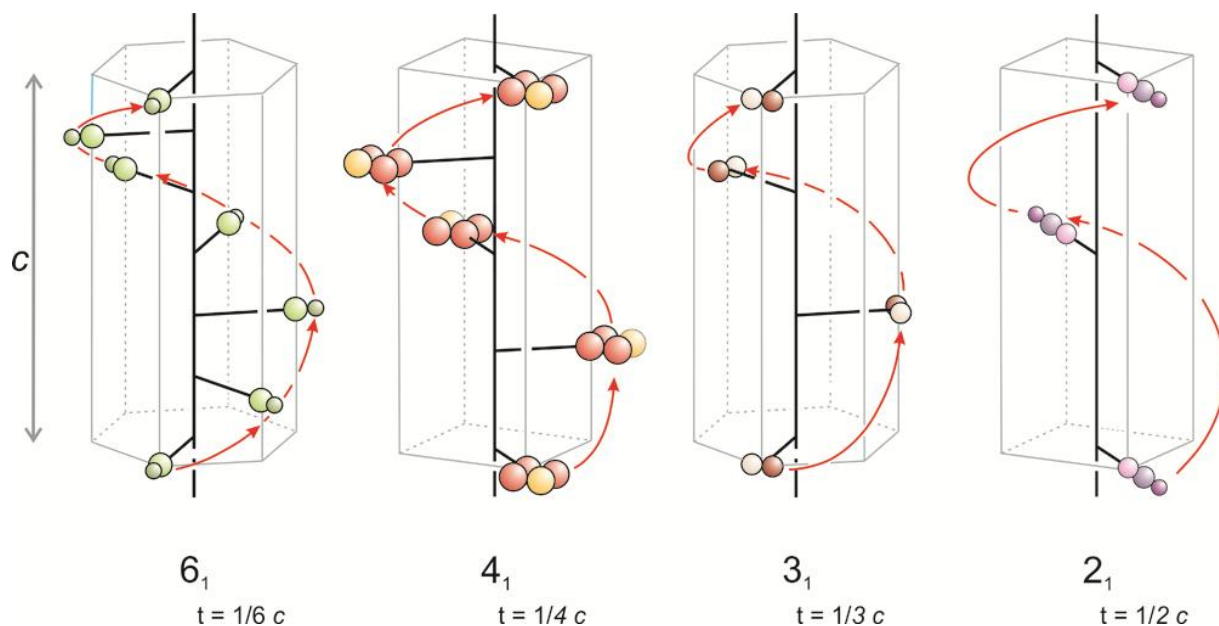


32个点群

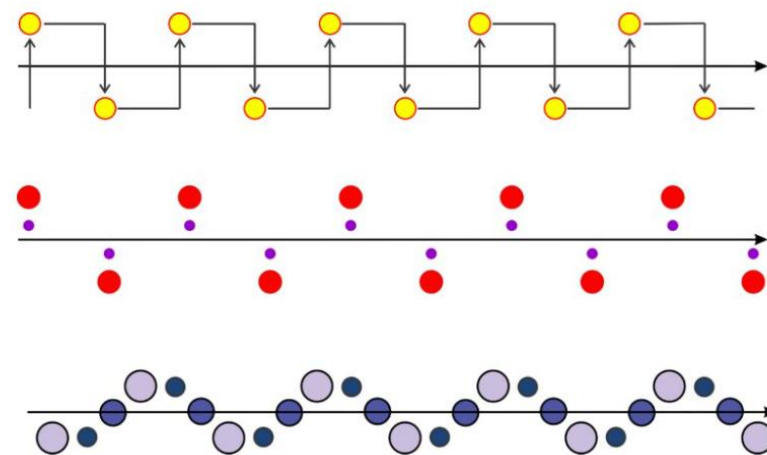
晶体的空间群操作

<https://opengeology.org/Mineralogy/11-crystallography/>

- 布拉维点阵角度：平移
- 点群角度：旋转、反演、镜面反射
- 平移复合点群操作
 - 螺旋轴 (screw axis)：平移 + 旋转



- 滑移面 (glide plane)：平移 + 镜面反射



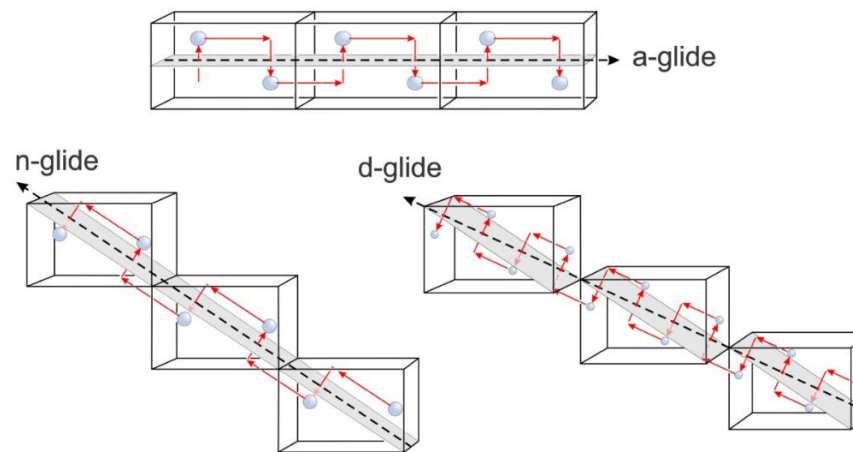
与转轴有关的20种 空间对称操作



Space Symmetry Operators Involving Rotation			
operator	type of operation	rotation angle	translation*
1 rotation axis	identity	360°	none
$\bar{1}$ rotoinversion axis	inversion center	360°	none
2 rotation axis	proper 2-fold	180°	none
2_1 screw axis	2-fold screw	180°	$1/2 t$
$\bar{2}$ rotoinversion axis	mirror	180°	none
3 rotation axis	proper 3-fold	120°	none
3_1 screw axis	3-fold screw	120°	$1/3 t$
3_2 screw axis	3-fold screw	120°	$1/3 t$
$\bar{3}$ rotoinversion axis	3-fold rotoinversion	120°	none
4 rotation axis	proper 4-fold	90°	none
4_1 screw axis	4-fold screw	90°	$1/4 t$
4_2 screw axis	4-fold screw	90°	$2/4 = 1/2 t$
4_3 screw axis	4-fold screw	90°	$3/4 t$
$\bar{4}$ rotoinversion axis	4-fold rotoinversion	90°	none
6 rotation axis	proper 6-fold	60°	none
6_1 screw axis	6-fold screw	60°	$1/6 t$
6_2 screw axis	6-fold screw	60°	$2/6 = 1/3 t$
6_3 screw axis	6-fold screw	60°	$3/6 = 1/2 t$
6_4 screw axis	6-fold screw	60°	$4/6 = 2/3 t$
6_5 screw axis	6-fold screw	60°	$5/6 t$
$\bar{6}$	6-fold rotoinversion	60°	none

* t = the unit cell dimension in the direction of translation

与镜面反射有关的 6种空间对称操作



Space Symmetry Operators Involving Reflection			
operator	type of operation	orientation of translation	translation*
<i>m-glide plane</i>	proper mirror	none	none
<i>a-glide plane</i>	axial glide	parallel to a	$\frac{1}{2} a$
<i>b-glide plane</i>	axial glide	parallel to b	$\frac{1}{2} b$
<i>c-glide plane</i>	axial glide	parallel to c	$\frac{1}{2} c$
<i>n-glide plane</i>	diagonal glide	par. to a face diagonal	$\frac{1}{2} t$
<i>d-glide plane</i>	diamond glide	par. to main diagonal	$\frac{1}{4} t$

*a, b, c, and t are the unit cell dimension in the direction of translation

不完全准确，
见下页

[2] An ***n*-glide** involves reflection then translation by half a unit cell in two axial directions: $(a+b)/2$, $(a+c)/2$, or $(b+c)/2$. An *n*-glide in three directions, $(a+b+c)/2$, is possible only for tetragonal and cubic lattices.

[3] **Diamond glides**, *d*, involve reflection then translation by one quarter of a unit cell in two axial directions. This glide operation is named for its presence in the crystal structure of diamond. The translation may be $(a+b)/4$, $(a+c)/4$, or $(b+c)/4$. A *d*-glide in three directions, $(a+b+c)/4$, is possible for cubic and tetragonal systems only.

空间群符号

例如金红石 (rutile) : $P4_2/m2_1/n2/m$

TABLE 6 The 230 Possible Space Groups

Crystal System	Point Group	Possible Space Groups
triclinic	1	$P1$
	$\bar{1}$	$P\bar{1}$
monoclinic	2	$P2, P2_1, C2$
	m	Pm, Pc, Cm, Cc
	$2/m$	$P2/m, P2_1/m, C2/m, P2/c, P2_1/c, C2/c$
orthorhombic	222	$P222, P222_1, P2_12_12, P2_12_12_1, C222_1, C222, F222, I222, I2_12_12_1$
	$mm2$	$Pmm2, Pmc2_1, Pcc2, Pma2, Pca2_1, Pnc2, Pmn2_1, Pba2, Pna2_1, Pnn2, Cmm2, Cmc2_1, Ccc2, Amm2, Abm2, Ama2, Aba2, Fmmc, Fdd2, Imm2, Iba2, Ima2$
	$2/m2/m2/m$	$P2/m2/m2/m, P2/n2/n2/n, P2/c2/c2/m, P2/b2/a2/n, P2_1/m2/m2/a, P2/n2_1/n2/a, P2/m2/n2_1/a, P2_1/c2/c2/a, P2_1/b2_1/a2/m, P2_1/c2_1/c2/n, P2/b2_1/c2_1/m, P2_1/n2_1/n2/m, P2_1/m2_1/m2/n, P2_1/b2/c2_1/n, P2_1/b2_1/c2_1/a, P2_1/n2_1/m2_1/a, C2/m2/c2/m, C2/m2/c2_1/a, C2/m2/m2/m, C2/c2/c2/m, C2/m2/m2/a, C2/c2/c2/a, F2/m2/m2/m, F2/d2/d2/d, I2/m2/m2/m, I2/b2/a2/m, I2/b2/c2/a, I2/m2/m2/a$

自学内容:

一般点 (general point)

特殊点 (special point)

等效点系: 由对称性关联的点

中文教材P253-256

Schönflies记号与国际记号的对照

中文教材P247

注意 $S_6 = C_{3i} = \bar{3}$

SCHOENFLIES	HEXAGONAL	TETRAGONAL	TRIGONAL	ORTHO-RHOMBIC	MONOCLINIC	TRICLINIC	INTER-NATIONAL
C_n	C_6  6	C_4  4	C_3  3		C_2  2	C_1  1	n
C_{nv}	C_{6v}  $6mm$	C_{4v}  $4mm$	C_{3v}  $3m$	C_{2v}  $2mm$			nmm (n even) nm (n odd)
C_{nh}	C_{6h}  $6/m$	C_{4h}  $4/m$			C_{2h}  $2/m$		n/m
	C_{3h}  $\bar{6}$				C_{1h} ($\bar{2}$)  m		$\bar{n}$
S_n		S_4  $\bar{4}$	S_6  (C_{3i}) $\bar{3}$			S_2  (C_i) $\bar{1}$	
D_n	D_6  622	D_4  422	D_3  32	D_2  (V) 222			$n22'$ (n even) $n2$ (n odd)
D_{nh}	D_{6h}  $6/mmm$	D_{4h}  $4/mmm$		D_{2h} (nmm)  (V_h) $2/mmm$			$\frac{n}{m} \frac{2}{m} \frac{2}{m}$ (n/mmm)
	D_{3h}  $\bar{6}2m$						$\bar{n}2m$ (n even)
D_{nd}		D_{2d}  (V_d) $\bar{4}2m$	D_{3d} ($\bar{3}m$)  $\bar{3} \frac{2}{m}$				$\bar{n} \frac{2}{m}$ (n odd)

问题三 晶胞中的点线面

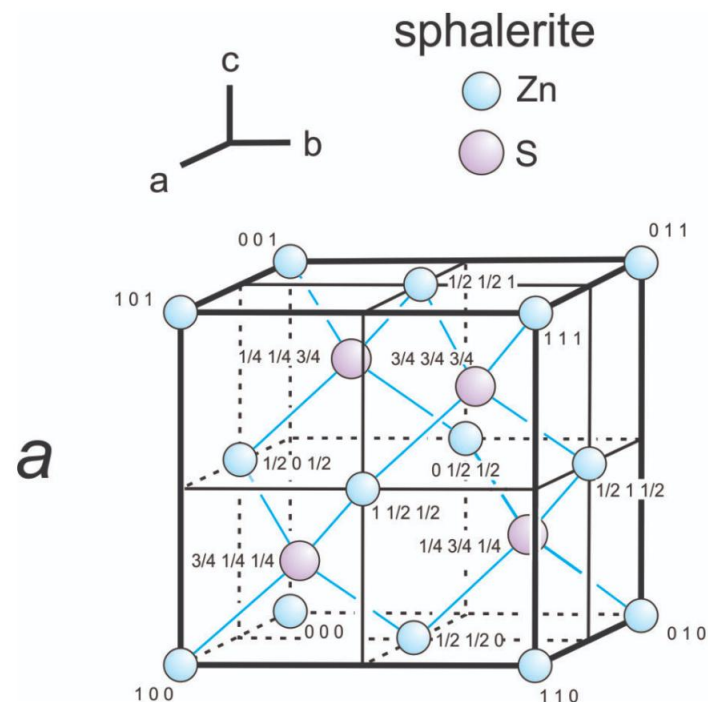
晶胞中的点

立方ZnS晶胞的

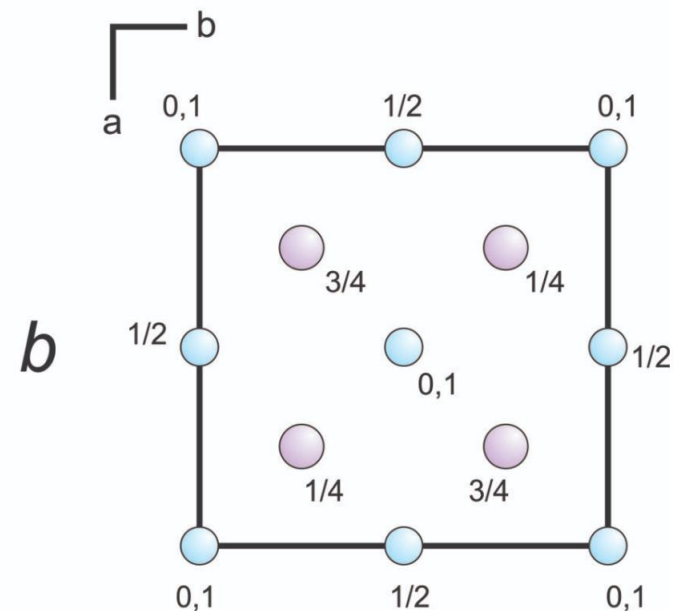
原子分数坐标

不加括号，如

$$\frac{1}{2} \frac{1}{2} 0; \frac{1}{4} \frac{1}{4} \frac{3}{4}$$



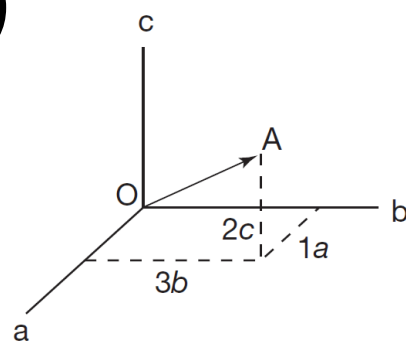
11.62 The atoms in a sphalerite unit cell



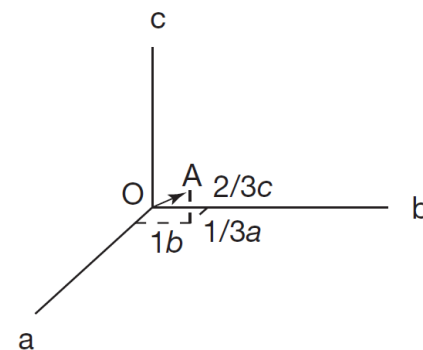
晶胞中的线（方向）

加中括号，如

[132]



(a) direction [132]



(b) direction $[\frac{1}{3} \ 1 \ \frac{2}{3}]$

垂直c方向投影图

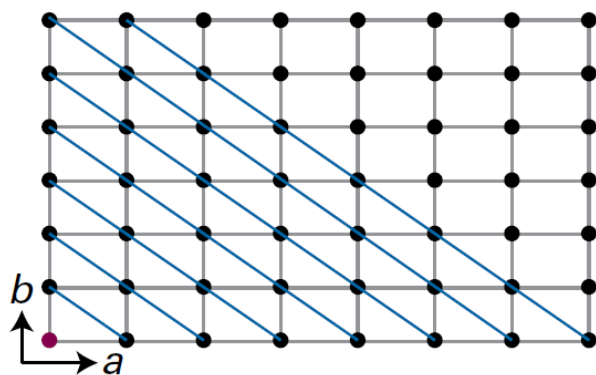
空隙

配位多面体

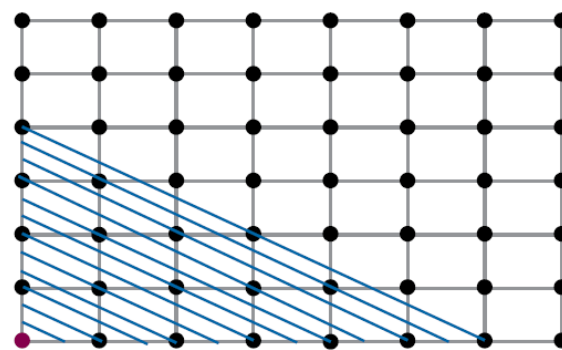
连接方式

晶面

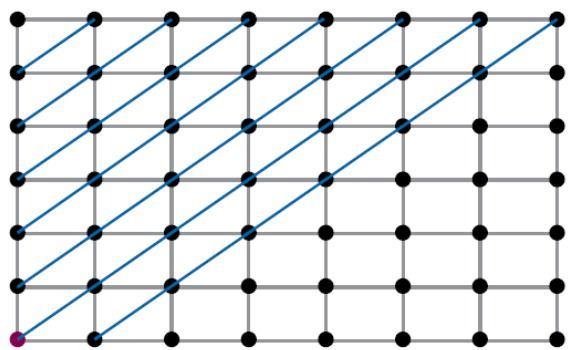
以二维点阵为例，显示晶面截距



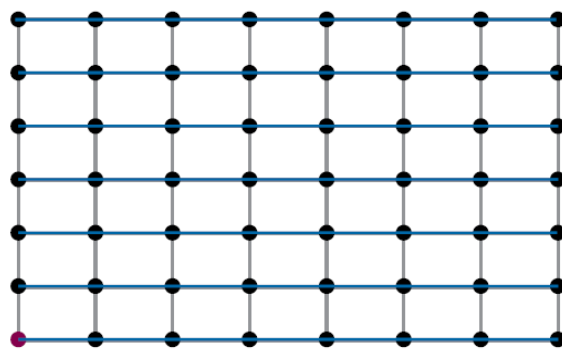
(a) $(1a, 1b)$



(b) $(\frac{1}{2}a, \frac{1}{3}b)$



(c) $(-1a, 1b)$



(d) $(\infty a, 1b)$

Miller指标:

截距取倒数,

再约化为互质整数 $(h\ k\ l)$

(a) (110)

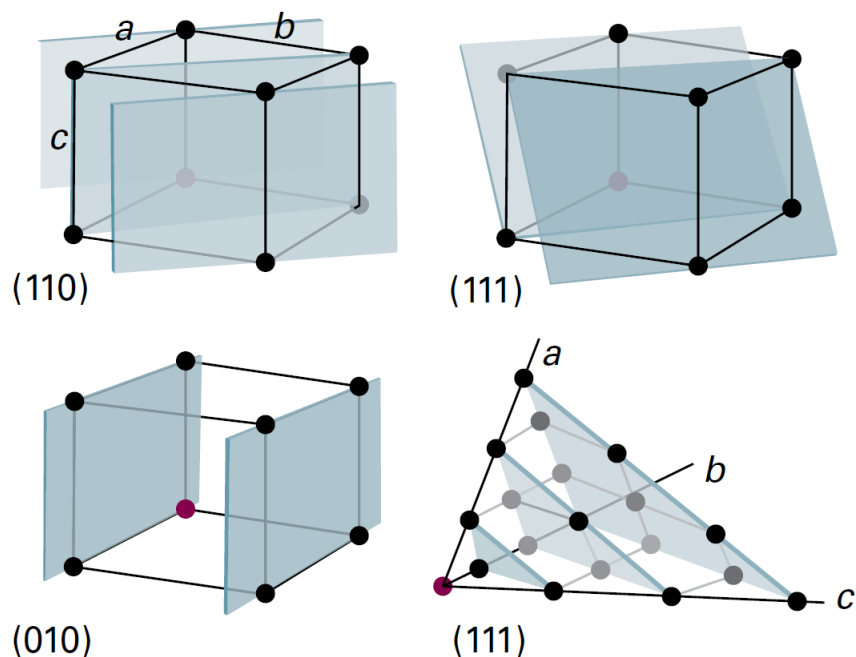
(b) (230)

(c) $(\bar{1}10)$

(d) (010)

某指标越小，越与相应
轴接近平行

指标为零，与轴平行



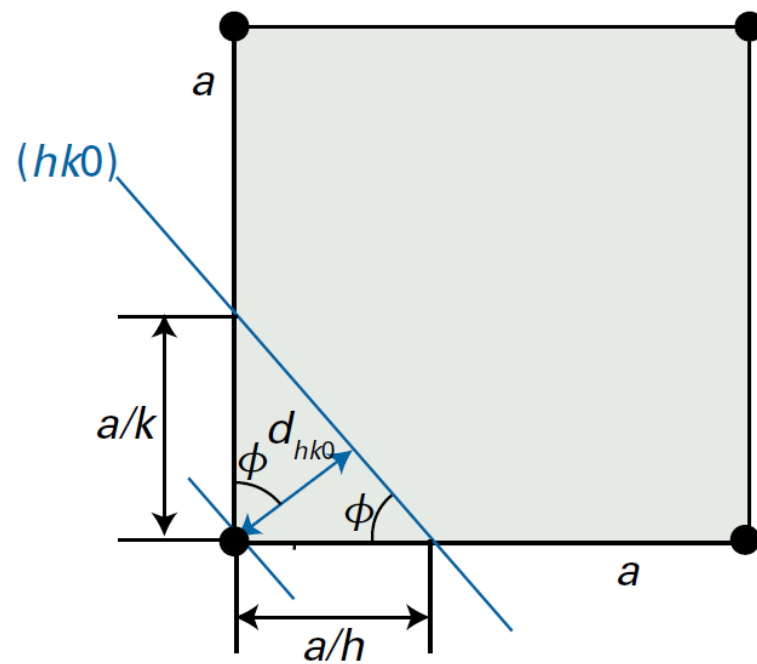
推广到三维晶面 (hkl)

立方点阵
$$d_{hkl} = \frac{a}{(h^2 + k^2 + l^2)^{1/2}}$$

正交点阵
$$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

晶面间距

以二维正方晶面 $(hk0)$ 为例



$$d_{hk0} = \frac{a}{(h^2 + k^2)^{1/2}}$$

参考教材

- P. Atkins, Physical Chemistry, 11th Edition, 2018.
- Neil W. Ashcroft, N. David Mermin, Solid State Physics, Cengage Learning, 1976.
- 结构化学基础, 周公度, 段连运, 第5版, 2017.
- Dexter Perkins, Mineralogy, 3rd edition, Pearson, 2011.
- <https://opengeology.org/Mineralogy/11-crystallography/>